



Cerise

**CENTRO DE EXCELÊNCIA EM REDES INTELIGENTES
SEM FIO E SERVIÇOS AVANÇADOS - UFG**

Introdução ao Conceito de Gateway 5G: Intercomunicação das Tecnologias Sem Fio

Grupo/Eixo Temático: 5G /O-RAN

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- **Algoritmos Tradicionais:** Implementação de métodos como MUSIC, ESPRIT, SEAD e MODEX.
- **Modelo de Sinal:** Sinais sintéticos gerados para geometrias, espaçamento e ruído customizáveis.
- **Visualização:** Apresenta ferramentas de geração de figuras baseadas na configuração escolhida.
- **Interfaces CLI e API:** Para interação com o simulador são fornecidas duas abordagens conforme conviniência.

Direction of Arrival Estimation: A Comprehensive Review and Open-Source Python Implementation

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Abstract—Direction of Arrival (DOA) estimation is a fundamental problem in array signal processing with applications in radar, sonar, wireless communications, acoustic source localization, and many other fields. This paper presents a comprehensive review of classical and modern DOA estimation algorithms, focusing on subspace-based methods. We discuss four main algorithm families: Multiple Signal Classification, Estimation of Signal Parameters via Rotational Invariance Techniques, Search of Directions by Differential Spectrum, and Method of Direction Estimation and its variation with extra roots. For each algorithm, we present the mathematical foundation, variants, advantages, limitations, and practical considerations. Furthermore, we contribute an open-source Python implementation of these algorithms, designed with a modular architecture that allows for extension and comparison. Our simulation framework enables researchers and practitioners to evaluate algorithm performance under various signal conditions, including varying SNR levels, angular separation, and source correlation. Extensive experimental results demonstrate the comparative strengths of each method, we compare the performance in challenging scenarios. The provided software package serves as both an educational resource and a practical tool for the signal processing community.

Index Terms—Direction of arrival (DOA) estimation, array signal processing, MUSIC, ESPRIT, SEAD, MODEX, Python

limitations have motivated the development of high-resolution methods that can provide more precise DOA estimates even under challenging conditions [4].

Subspace-based methods have emerged as powerful approaches for DOA estimation, offering superior resolution compared to conventional techniques. Among these, the Multiple Signal Classification (MUSIC) algorithm and its variants have been widely adopted due to their ability to resolve closely spaced sources [5]. The Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT) algorithm presents an alternative approach that avoids the exhaustive search required by MUSIC, offering computational advantages [6]. More recent methods, such as Search of Directions by Differential Spectrum (SEAD) and Method of Direction Estimation (MODE) and MODE with extra roots (MODEX), address specific challenges including coherent signal environments and further improvements in resolution capability [7], [8].

Despite the theoretical advances in the field, practical implementation and comparative evaluation of these algorithms present significant challenges. Researchers and practitioners

Obrigado!